PROJECT FLOW DOCUMENT

Malaria Detection Using Deep Learning

Project Title: Malaria Detection Using Deep Learning
Technology Used: Python, TensorFlow, OpenCV
Model Type: Convolutional Neural Network (CNN)

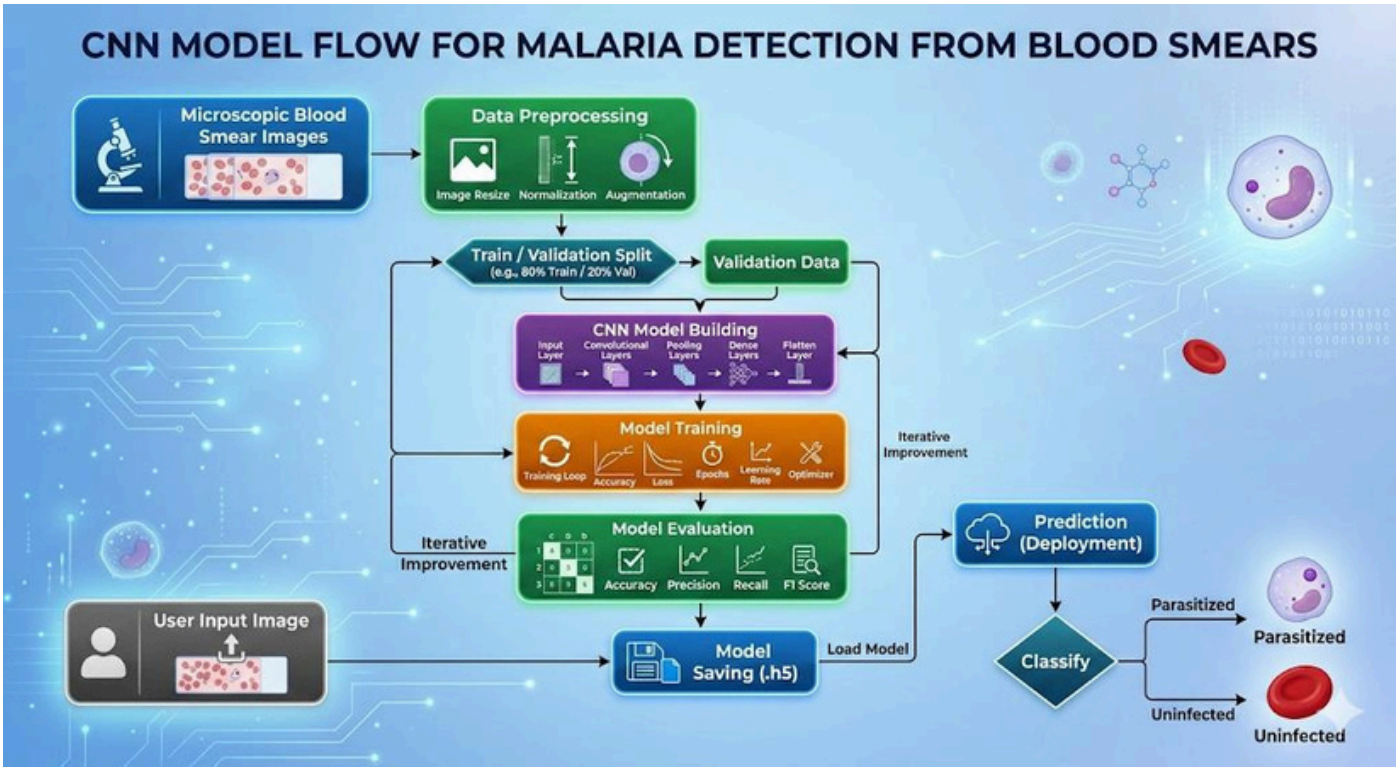
Introduction

Malaria is a life-threatening disease caused by parasites transmitted through mosquito bites. Early and accurate detection is essential for effective treatment.

This project uses Deep Learning techniques to automatically classify microscopic blood smear images as **Parasitized (Infected)** or **Uninfected (Healthy)** using a Convolutional Neural Network (CNN).

Project Flow Overview

The system follows a structured pipeline from data collection to final prediction.



Detailed Project Flow Explanation

3.1 Dataset Collection

Microscopic blood smear images are collected and organized into two categories:

- **Parasitized** (infected red blood cells)
- **Uninfected** (healthy red blood cells)

The dataset is structured into folders for proper training and validation.

3.2 Data Preprocessing

Before training the model, images undergo preprocessing to ensure consistency and better model performance.

The preprocessing steps include:

- Resizing images to 64 × 64 pixels
- Normalizing pixel values (scaling from 0–255 to 0–1)
- Splitting dataset into training and validation sets

This step ensures uniform input size and improves model accuracy.

3.3 Train–Validation Split

The dataset is divided into:

- **Training Data** – Used to train the model
- **Validation Data** – Used to test model performance

This helps in evaluating how well the model generalizes to unseen data.

3.4 Model Building

A Convolutional Neural Network (CNN) model is built using TensorFlow and Keras.

The architecture includes:

- Convolutional layers for feature extraction
- Max pooling layers for dimensionality reduction
- Fully connected (Dense) layers
- Sigmoid activation function for binary classification

The model is compiled using:

- Optimizer: Adam
- Loss Function: Binary Crossentropy
- Metric: Accuracy

3.5 Model Training

The CNN model is trained using the training dataset.

During training:

- The model learns patterns in infected and uninfected cells
- Accuracy and loss are monitored
- Validation data prevents overfitting

The training process continues for multiple epochs until optimal performance is achieved.

3.6 Model Evaluation

After training, the model is evaluated using validation data.

Performance metrics include:

- Accuracy
- Precision
- Recall
- Confusion Matrix

This step ensures the model performs reliably before deployment.

3.7 Model Saving

After successful training and evaluation, the model is saved as:

malaria_model.h5

This file stores the trained weights and architecture and is later used for prediction.

3.8 Prediction Module

The prediction module allows users to input a new microscopic image.

The process includes:

1. Readingtheimage using OpenCV
2. Resizingandnormalizing it
3. Feedingitintothe trained model
4. Generatingprediction

Output is displayedas:

- **Parasitized**
  or
- **Uninfected**

Final Working of the System

The Malaria Detection System is deployed as a simple web-based application that allows users to :

1. User Uploads Image: The user opens the website and uploads a microscopic blood smear image through the user interface.
2. Image Preprocessing : The uploaded image is automatically resized and normalized to match the input format required by the trained deep learning model.
3. Model Prediction : The preprocessed image is passed to the trained Convolutional Neural Network (CNN) model, which analyzes the image features.
4. Classification Result : The model predicts whether the blood cell image is Parasitized (infected) or Uninfected (healthy).
5. Result Display : The prediction result is displayed on the website interface, allowing the user to quickly understand the diagnosis.

Conclusion

The Malaria Detection System successfully uses Deep Learning techniques to classify blood smear images as infected or uninfected.

The project demonstrates how artificial intelligence can assist in medical diagnosis by providing fast, accurate, and automated detection of malaria.